

# Leveraging Generative AI and Prompt Engineering to Enhance Cloud Operations

We explore the benefits of using generative AI and prompt engineering for cloud operations. Popular GenAI tools for major cloud platforms and cloud-agnostic solutions are also listed, along with real-world examples of effective prompt engineering.



**G**enerative AI is revolutionising cloud operations by providing intelligent solutions for monitoring, management, and troubleshooting. By leveraging prompt engineering, professionals can create scripts that automate various operational tasks, enhancing efficiency and reducing human error. This is possible with ‘vibe coding’, where we can use a prompt to generate computer-assisted coding for cloud operations and cloud service management.

## Prompt engineering for cloud operational services

Prompt engineering involves crafting effective input prompts to elicit the desired responses from generative AI models. In the realm of cloud operations, it can be used to automate tasks such as monitoring system performance, managing resources, and troubleshooting issues. Here’s how prompt engineering can be used across different operational services.

**Cloud monitoring:** For L1 operations, generative AI can be prompted to continuously monitor cloud infrastructure, detecting anomalies and performance bottlenecks. Scripts can be designed to check resource utilisation, identify latency issues, and alert operators to potential security threats.

**Cloud management:** L2 operations benefit from generative AI by automating resource allocation, scaling, and configuration management. Prompt engineering can